Xiangyu Wang

Undergraduate · School of Life Science and Technology · ShanghaiTech University

I am an undergraduate student in Biological Sciences at ShanghaiTech University, currently on exchange at UC Berkeley. My interests span biochemistry, molecular biology, and the intersection of biology with computational tools.

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Education

2023 – 2027 **ShanghaiTech University**
B.S. in Biological Sciences · School of Life Science and Technology (SLST).

Spring 2026 **University of California, Berkeley**
Exchange Student · Berkeley Biosciences Study Abroad (BBSA) Program.

Awards & Honors

2025 **2025 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM 2025)**
National Second Prize and Shanghai Regional First Prize.

Research & Lab Experience

2024 – Present **Sun Yadong Lab, ShanghaiTech University (SLST)**
Undergraduate Researcher
Undergraduate researcher in the Sun Yadong Lab, studying the structure and function of protein-RNA complexes using structural biology techniques like cryo-electron microscopy. Gained expertise in plasmid construction, cell transfection, protein expression and purification.

Fall 2025 **FlyCropper – Drosophila Wing Classification System**
Genetics Lab Independent Project
One of the primary code contributors for an independent experimental project in the Genetics laboratory course. The system uses a two-stage deep-learning pipeline (object detection + classification) to automatically detect Drosophila and classify wing phenotypes (long / short wing).
github.com/xiangyuw1/FlyCropper

Spring 2025 **Evaluating Inhibitory Effects of mTORC1 Inhibitors on Cancer Cell Lines**
Cell Biology Lab Independent Project
Responsible for the design and execution of the biochemistry experimental component of this independent project in the Cell Biology laboratory course, investigating the inhibitory effects of mTORC1 inhibitors on multiple cancer cell lines.

Fall 2025 **Computational Simulation of Circadian Rhythm Regulation in Cyanobacteria**
Introduction to Systems Biology Course Project
Built and analyzed a core KaiA/KaiB/KaiC oscillator model: KaiA promotes KaiC phosphorylation while KaiB provides antagonistic negative feedback, causing cyclical shifts in KaiC phosphorylation state that produce an approximately 24-hour circadian rhythm.

Other Projects

- **Chi_Square**
  A comprehensive Python chi-square calculator covering the majority of statistical computation problems encountered in genetics courses, including goodness-of-fit tests, contingency table tests (with optional Yates' correction), and Hardy-Weinberg equilibrium tests. Inspired by and originally initiated by github.com/SiyuanMa42.
  `github.com/xiangyuw1/chi_square`

Selected Coursework

- Linear Algebra I — A
- Introduction to Information Science & Technology — A+
- Biochemistry I — A
- Fundamentals of Organic Chemistry — A
- Introduction to Systems Biology — A+
- Light Microscopy — A+
- Brain Organoids — A
- Genetics — A

Skills

- Molecular Biology (wet lab)
- Biochemistry (wet lab)
- Vibe Coding

Languages

- Chinese: Native
- English: Duolingo English Test 115; CET-6 524